

Oasis Ablation Appendix

Sections 4.3, 4.4, 4.5, 5.2, 5.3 (+ related stack ablations)

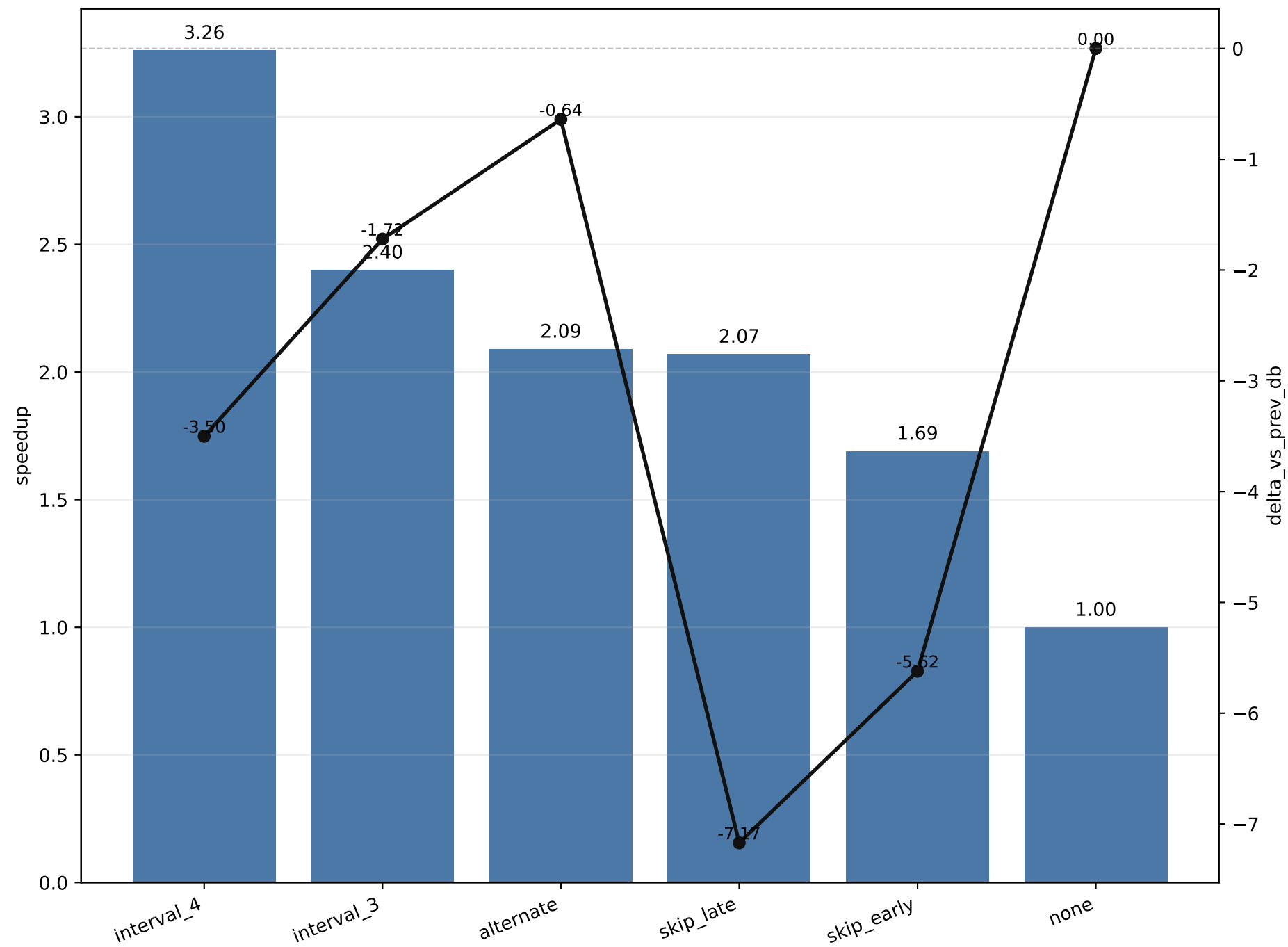
Source: `benchmarks/results/all_runs.csv`

Section 4.3 — Step Caching Patterns (32 frames)

variant	config	speedup	delta_vs_prev_db	cross_psnr_db	verdict
interval_4	75% skip [10 6 2]	3.26	-3.50	15.32	quality_breaking
interval_3	67% skip [10 7 4 1]	2.40	-1.72	18.78	stable
alternate	50% skip [10 8 6 4 2]	2.09	-0.64	19.99	stable_at_short
skip_late	clustered low noise	2.07	-7.17	9.61	catastrophic
skip_early	clustered high noise	1.69	-5.62	15.49	catastrophic
none	baseline	1.00	0.00	—	reference

Takeaway: distribution of skips matters more than raw skip rate.

Section 4.3 — Step Caching: Speedup vs Coherence Delta (32 frames)



Section 4.3 — Step Caching Length Validation (75/950 frames)

variant	length_frames	speedup	delta_vs_prev_db	verdict	notes
alternate	950	1.94	-6.20	degrades_at_length	blur compounds
interval_3	75	2.49	-0.17	stable	—
interval_3	950	2.47	-1.56	stable	—

Section 4.4 — DPM-Solver++ 2M Results

variant	length_frames	speedup	delta_vs_prev_dc	cross_psnr_db	verdict	notes
DPM-Solver++ 2M	32	2.21	2.03	20.31	stable	smoothing-as-regularization
DPM-Solver++ 2M	950	1.98	4.55	12.95	stable	—

Section 4.5 — Difficulty Schedule Buckets / Formula Variants (32 frames)

	variant	config	speedup	delta_vs_prev_dc	cross_psnr_db	verdict	notes
aggressive	r=2/m=4/	thr 0.5/1.5	2.44	0.76	21.12	winner	45% step savings
	linear slope=2.5	smooth formula	2.17	1.07	21.11	worse_than_bucket	—
super-agg	r=1/m=3/	thr 0.5/1.5	2.10	1.34	17.76	cliff	reduced=1 cliff
exponential	decay=	smooth formula	1.77	1.19	22.16	best_quality_low_speed	—
	default	r=4/m=6/f=	1.74	1.38	19.64	decent	32.5% step savings
conservative	r=8/m=9	thr 0.5/1.5	1.16	0.53	22.26	marginal	10% step savings

Section 4.5 — Difficulty Length Validation (950 frames)

variant	length_frames	speedup	delta_vs_prev_db	verdict	notes
aggressive r=2/m=4/f=	950	2.69	0.76	winner	length-robust 61.5% saving

Section 5.2 — Stack: DPM++5 + Alternate Stepcache

variant	length_frames	speedup	delta_vs_prev_dc	cross_psnr_db	verdict	notes
++5 + stepcache_a	32	3.35	0.68	—	blur_visible	metric fooled by smoothing
++5 + stepcache_a	950	3.35	-6.72	—	blur_compact	metric agrees with visual

Section 5.3 — Stack: DPM++5 + Difficulty (Headline Pair)

variant	length_frames	speedup	delta_vs_prev_dc	cross_psnr_db	verdict	notes
+5 + Difficulty r=2,	32	3.05	2.22	20.58	clean_win	—
+5 + Difficulty r=2,	950	3.54	-0.02	11.23	project_head9500	9500 to 2683 forwards

Related Stack Ablation — DPM++5 + Difficulty + TaylorSeer (Megastack)

variant	length_frames	speedup	delta_vs_prev_dc	cross_psnr_db	verdict	notes
stack DPM++5 + Difficulty	32	3.52	1.45	15.95	acceptable	Extrapolates from TaylorSeer on top of DP
stack DPM++5 + Difficulty	950	3.76	-11.29	8.05	predictable	length drops to 8.8% lands on f=5 high-

Included for presentation Q&A since it degrades strongly at long horizon.

Additional Not-on-Poster Ablations (Compile / Predit / Drift / Action-KV / Warmstart)

family	variant	length_frame	speedup	delta_vs_prev	cross_psnr_dl	verdict	notes
action_kv	sim=0.85 loose	32	1.64	-3.43	18.48	degraded	24/32 frames qualify
action_kv	sim=0.95 default	32	1.55	-3.43	18.48	degraded	24/32 frames qualify
action_kv	sim=0.99 strict	32	1.41	-3.52	18.88	degraded	14/32 frames qualify
action_kv	sim=0.95 default	950	1.58	-21.14	—	catastrophic	length compounding
drift_cache	buggy consecutive	32	9.33	-30.00	7.27	buggy	90% skip garbage output
drift_cache	tau=0.05 strict	32	1.47	-0.53	19.13	stable	formula validated 30.9% skip
drift_cache	tau=0.10 default	32	1.86	-0.94	20.53	stable	45.3% skip
drift_cache	tau=0.20 loose	32	2.26	-1.17	18.75	stable_at_short	54.7% skip
drift_cache	tau=0.20 loose	950	2.26	-4.36	—	degrades_at_length	4x quality drop
predit	AB-2 alpha=1.5	32	1.44	—	—	worse_than_predit	40% predict rate due to history needs
torch_compile	default + clear rotary	32	—	—	—	failed	attribute error
torch_compile	mode=default fixed	32	1.08	—	—	marginal	that works only no CUDA graphs
torch_compile	reduce-overhead	32	—	—	—	failed	hung in recompile loop
torch_compile	mode=reduce-overhead	75	—	—	—	empty CUDA graphs	from rotary graph break
warmstart	k=6 blend=0.5	32	1.69	19.26	—	frozen	severe stutter
warmstart	k=8 blend=0.0	32	1.23	2.54	—	looks_fine_at_short	—
warmstart	k=8 blend=0.2	32	1.26	18.36	—	frozen	still frozen
warmstart	k=8 blend=0.0	950	1.25	11.91	—	frozen_at_length	—